GridPilot: Intelligent Orchestration of EV Charging to Prevent Grid Bottlenecks and Reduce Carbon Emissions

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ABSTRACT

India is in the middle of one of the fastest electric vehicle adoptions in the world. Corporate fleets, logistics operators, and ride-hailing companies are deploying hundreds of EVs simultaneously — yet the electricity infrastructure supporting them was never designed for this scale. Every night, when large EV fleets return to their depots and plug in together, they create a sudden, simultaneous surge in electricity demand that overwhelms local transformers, triggers costly penalty charges from distribution companies, and draws power from the dirtiest, most carbon-heavy part of the grid. No software platform in India currently solves this problem intelligently. GridPilot is that platform. It is a grid-edge EV charging orchestration system that coordinates when each vehicle in a fleet charges, how much power it draws, and which grid windows to target — using a mathematical optimizer running on a convex quadratic program that solves in under four seconds for a 500-vehicle fleet. Instead of all 500 vehicles charging simultaneously and pushing the transformer to 185% of rated capacity, GridPilot staggers and shapes the load, shifts charging toward the cleanest hours of the night, and guarantees every vehicle is fully charged before the morning shift. Modeled on the operational profile of Lithium Urban Technologies, India's largest corporate electric fleet operator in Gurugram, GridPilot demonstrated a 51.2% reduction in peak transformer load, elimination of all five nightly overload events, an 11.9% reduction in charging-related carbon emissions, and monthly electricity savings of ₹7.35 lakh per depot — delivered entirely through software, with no hardware changes required.

Objectives

The rapid electrification of commercial vehicle fleets in urban India presents a critical infrastructure challenge: uncoordinated overnight EV charging creates severe transformer overloads, peak demand penalties, and elevated carbon emissions at depot-scale facilities. This paper presents GridPilot, a grid-edge EV charging orchestration platform designed to prevent distribution infrastructure bottlenecks while minimizing the carbon intensity of large fleet charging operations. The platform is modeled on the operational profile of Lithium Urban Technologies, India's largest corporate electric fleet operator based in Gurugram, Haryana — a facility connected to the DVVNL HT-2 grid with no intelligent charging orchestration currently deployed. The primary objectives are to eliminate transformer overload events through real-time load scheduling, reduce charging-related carbon emissions by shifting demand to low-intensity grid windows, and demonstrate financial viability through quantified DVVNL demand charge reduction.

Methods

GridPilot employs a convex quadratic programming scheduler implemented using CVXPY with the ECOS solver, minimizing a weighted objective function across four terms: carbon cost ($\alpha = 0.35$), transformer peak penalty ($\beta = 0.35$), user discomfort ($\gamma = 0.20$), and DVVNL demand charge exposure ($\delta = 0.10$). The scheduling horizon spans 15-minute intervals across an overnight charging window, with 500 vehicles modeled as decision variables subject to hard grid constraints. Charging behavior is modeled using session data adapted from the Caltech ACN-Data dataset (Flores-Espino et al., 2021), rescaled to reflect a 500-vehicle Tata Nexon EV fleet (40 kWh battery, 7.4 kW AC charger) operating under real DVVNL HT-2 tariff structures. Carbon intensity signals are derived from Central Electricity Authority state-wise emission factors (Haryana baseline: 0.820 kg CO_2/kWh; CEA, 2023). National grid stress signals are processed through a secondary intelligence layer, FirstFlight, which provides hourly carbon intensity forecasts consumed directly by the scheduling objective function. Campus-level power flow dynamics are validated through a pandapower physics simulation of a seven-bus depot microgrid, ensuring all optimizer recommendations satisfy physical grid constraints.

Results

Simulation results demonstrate a 51.2% reduction in transformer peak load, from an unmanaged peak of 4,100 kW — representing 185% of rated transformer capacity — to a GridPilot-scheduled peak of 2,000 kW, remaining safely within the 4,000 kW continuous operating limit throughout the overnight charging window. The unmanaged baseline produced five distinct overload events per night; the GridPilot schedule produced zero. Carbon emissions from fleet charging were reduced by 11.9%, saving 1,206 kg CO_2 per night, through demand shifting toward lower-intensity grid windows identified by the FirstFlight carbon signal layer. All 500 vehicles were confirmed ready at 80% state of charge by the 07:00 IST morning dispatch deadline. Monthly DVVNL demand charge savings were computed at ₹7.35 lakh per depot, derived from the 2,100 kW peak reduction applied against the DVVNL HT-2 demand charge rate of ₹350/kVA/month. The CVXPY optimizer converged to an optimal solution in 3,943 milliseconds for the full 500-vehicle fleet under local development conditions; solve time is expected to reduce substantially under production hardware with pre-warmed solver state. Demand forecasting models achieved a mean MAPE of 0.83% across all five national grid regions, and anomaly detection F1 scores averaged 0.95, both significantly exceeding pre-defined validation targets of 8% MAPE and 0.70 F1 respectively. These model metrics reflect performance on synthetically generated holdout data and may not directly generalise to live grid measurements.

Conclusion

GridPilot demonstrates that software-defined EV load orchestration, requiring no hardware modifications to existing charging or grid infrastructure, represents a viable and immediately deployable pathway toward grid-resilient, carbon-aware fleet electrification in India's commercial transport sector. The platform addresses a documented and growing infrastructure gap: as corporate EV fleet penetration accelerates across NCR and other Indian metros, depot-scale transformer stress will become a systemic barrier to adoption without intelligent orchestration. By combining convex optimization, real CEA carbon data, and physics-validated power flow simulation, GridPilot provides a technically rigorous and financially compelling solution that aligns grid reliability with decarbonization objectives. Future work will focus on hardware pilot deployment at an institutional fleet facility, integration with OCPP 2.0.1-compliant smart chargers, and extension of the Vehicle-to-Grid dispatch module to enable DISCOM demand response revenue.

Keywords: *EV charging optimization, smart grid, convex optimization, carbon-aware scheduling, India power distribution*

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